Experiment - 2

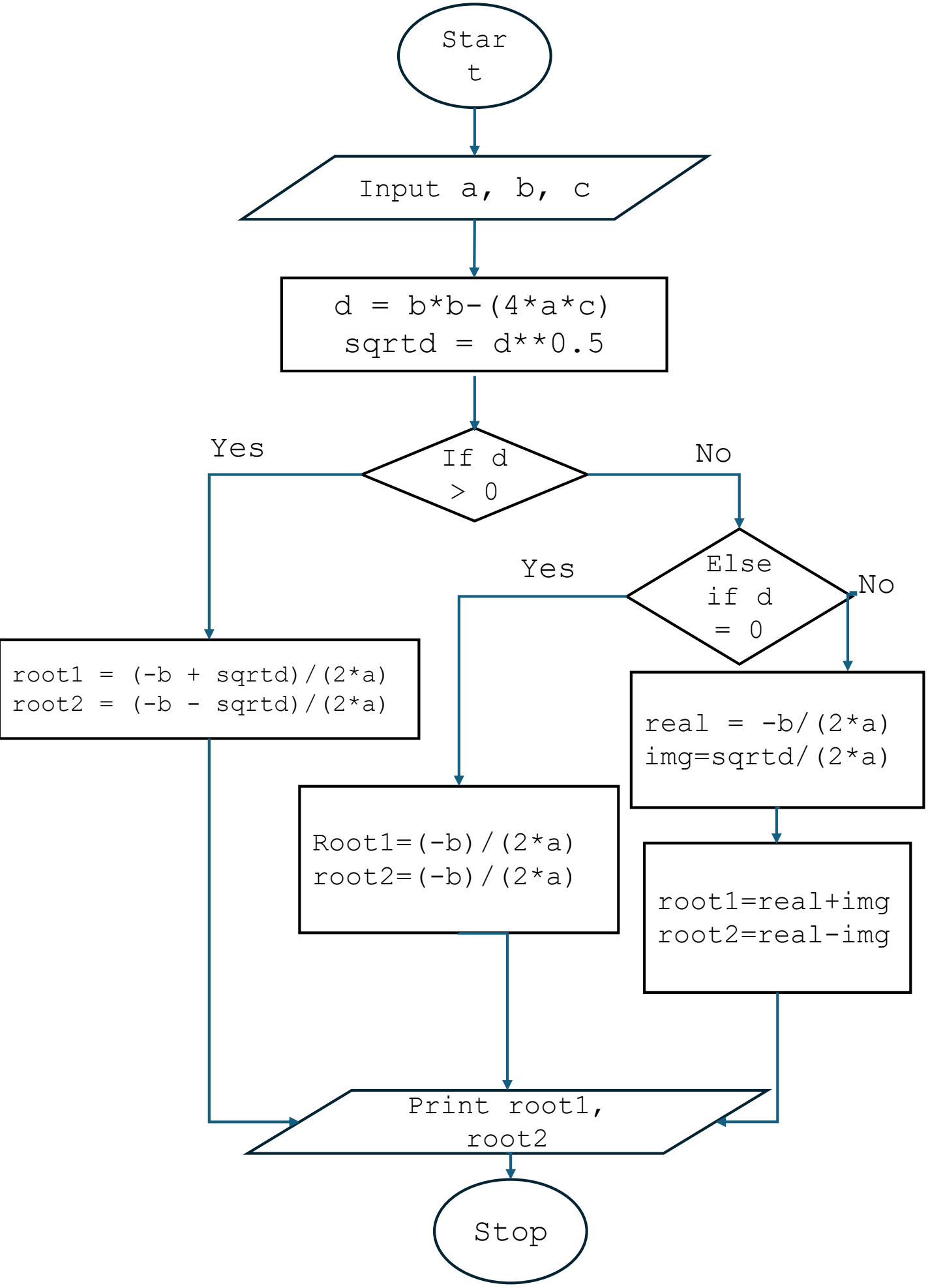
2.1.1 Roots of a Quadratic Equation

- Algorithm

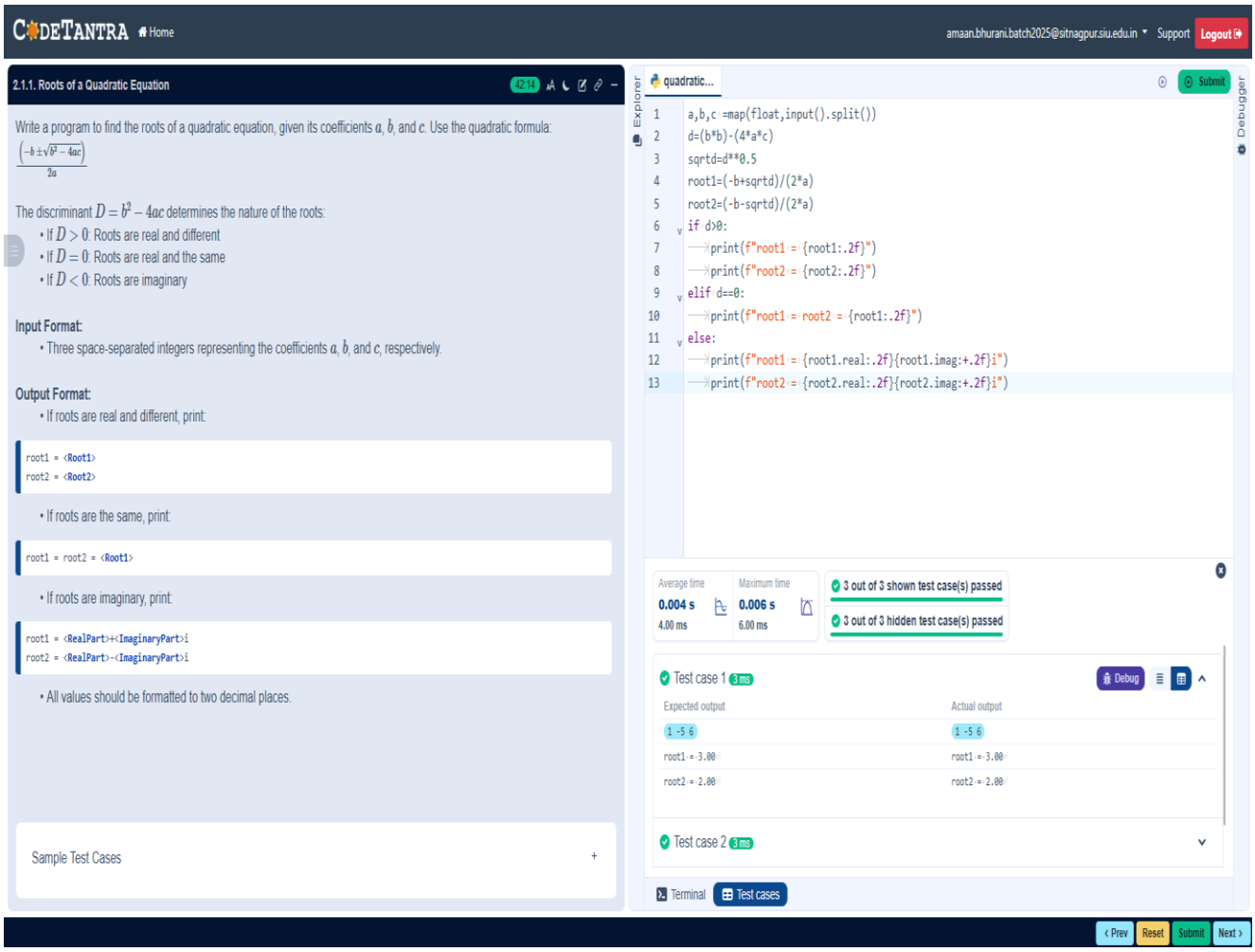
```
STEP 1 : Start
STEP 2 : Input  a, b, c
STEP 3 : Calculate
         D = b*b - 4*a*c
STEP 4 : if D > 0
            root1 = (-b+√D)/(2*a)
            root2 = (-b-√D)/(2*a)
         else if D==0
            root1 = root2 = (-b)/(2*a)
         else
            real = (-b)/(2*a)
            imaginary = (√-D)/(2*a)
            root1 = real + imaginary
            root2 = real - imaginary

STEP 5 : Print root1, root2
```

- Flowchart


Star
t
Input a, b, c
d = b*b-(4*a*c)
sqrtd = d**0.5
If d
> 0
Yes
No
Else
if d
= 0
Yes
No
root1 = (-b + sqrtd)/(2*a)
root2 = (-b - sqrtd)/(2*a)
real = -b/(2*a)
img=sqrtd/(2*a)
Root1=(-b)/(2*a)
root2=(-b)/(2*a)
root1=real+img
root2=real-img
Print root1,
root2
Stop

- Code

```
a,b,c =map(float,input().split())
d=(b*b)-(4*a*c)
sqrtd=d**0.5
root1=(-b+sqrtd)/(2*a)
root2=(-b-sqrtd)/(2*a)
if d > 0:
    print(f"root1 = {root1:.2f}")
    print(f"root2 = {root2:.2f}")
elif d==0:
    print(f"root1 = root2 = {root1:.2f}")
else:
print(f"root1 =
{root1.real:.2f}{root1.imag:+.2f}i")
print(f"root2 =
{root2.real:.2f}{root2.imag:+.2f}i")
```

- Execution